

IoT-Based Campus Router Health 360 and Digital Twin Fleet Simulator

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Smart Campus IoT Infrastructure • Standards-Compliant RFC 8428 & DTDL v2
Prototype

ABSTRACT

Modern university campuses and enterprise facilities depend heavily on reliable, high-density wireless and wireline network infrastructure. Campus edge router gateways form the vital backbone connecting tens of thousands of student laptops, laboratory workstations, and distributed Internet of Things (IoT) sensors including environmental monitors, smart energy meters, and security cameras. Traditional network management systems often rely on static SNMP polling, threshold alerts, and manual user trouble tickets, which frequently lead to unnoticed bufferbloat, thermal throttling, RF noise degradation, and delayed responses to critical hardware faults.

This project presents an enterprise-grade IoT-based Campus Router Health 360 and Digital Twin Fleet Simulator, engineered using FastAPI, modern React, and standards-compliant telemetry protocols. The system continuously simulates and monitors a mesh fleet of 42 virtual IoT edge router gateways distributed across 5 campus operational zones. The architecture incorporates standard IETF RFC 8428 Sensor Measurement Lists (SenML) for sensor telemetry serialization and the W3C Digital Twin Definition Language (DTDL v2) for unified device modeling.

Each simulated router gateway models realistic physical dynamics including SoC core temperature, PoE+ power consumption, compute load, RAM utilization, RF noise floor (-95 to -55 dBm), packet loss, and jitter. An interactive Chaos & Fault Injection Laboratory allows network administrators to simulate real-world failure modes such as thermal runaway ($>95^{\circ}\text{C}$), buffer memory leaks, RF jamming, and broadcast DDoS storms. The platform integrates an explainable machine learning pipeline utilizing XGBoost and SHAP feature attribution to detect degradation patterns up to 48 hours before physical service failure.

The project demonstrates how digital twin abstractions, standardized IoT telemetry, and real-time 60 FPS particle mesh visualization can transform reactive network troubleshooting into proactive, self-healing edge infrastructure management.

INTRODUCTION

The Internet of Things (IoT) provides a standardized framework to connect sensors, microcontrollers, edge gateways, and cloud analytics into cohesive automated monitoring ecosystems. In enterprise and educational environments, campus routers function not only as packet forwarders but as primary IoT edge gateways aggregating high-frequency sensor streams.

This project establishes a comprehensive IoT-based network health simulation platform named NetSentinel. By combining virtual edge router gateways, SenML data normalization, digital twin synchronization, and bi-directional edge actuation, the system enables complete end-to-end lifecycle observation without requiring expensive physical hardware deployments during development and testing.

PROBLEM STATEMENT

Traditional campus network operations rely on fragmented management consoles, fixed periodic polling intervals, and reactive user complaints. Such approaches fail to anticipate progressive hardware degradation and localized environmental stresses.

The major operational problems include:

- Unnoticed thermal throttling in unventilated campus wiring closets causing severe latency spikes.
- Silent buffer memory leaks in firmware resulting in sudden packet drops and interface flaps.
- Lack of real-time RF spectrum noise floor visibility across dynamic Wi-Fi 6 channels.
- Inability to safely test catastrophic failure modes without taking live production systems offline.
- Excessive technician dispatch times due to ambiguous root-cause diagnostic data.
- High financial cost and logistical difficulty of procuring dozens of physical test routers.
- Lack of standards-compliant digital twin models for automated remote edge actuation.

Therefore, there is an urgent need for an IoT-based simulator and digital twin platform that continuously monitors physical and network parameters, simulates anomalies, and executes proactive mitigations.

OBJECTIVES

The main objectives of this project are:

1. To design a standards-compliant IoT Edge Router Fleet Simulator using asynchronous Python services.
2. To encode all edge telemetry using the official IETF RFC 8428 SenML specification.
3. To model physical SoC core temperature dynamics and chassis thermal inertia.
4. To monitor active PoE+ power draw consumption and client device saturation.
5. To simulate RF spectrum background noise floor and dynamic channel interference.
6. To implement the W3C DTDL v2 Digital Twin interface for unified gateway modeling.
7. To construct an interactive Chaos & Fault Injection Laboratory for testing anomalies.
8. To integrate an XGBoost and SHAP explainability engine for predictive degradation scoring.
9. To build a 60 FPS HTML5 Canvas particle mesh visualizer showing real-time telemetry flow.
10. To provide remote bi-directional actuation controls including cryo-cooling and DFS channel hopping.
11. To deliver a production-ready full-stack web application accessible to campus network operators.

SCOPE OF THE PROJECT

The project focuses on developing an end-to-end IoT-based campus edge router simulation platform and digital twin operations deck.

The current project scope includes:

- 42 virtual digital twin router gateways across 5 distinct campus building zones.
- Real-time multi-parameter sensor telemetry generation (temperature, CPU, RAM, power, noise, loss).
- Standardized IETF RFC 8428 SenML JSON stream generation and serialization.
- W3C DTDL v2 schema contract definition with bidirectional actuation commands.
- Chaos fault injection engine supporting thermal runaway, memory leaks, RF jamming, and DDoS storms.
- Virtual 1U physical chassis faceplate with dynamic LED bank indicators (PWR, SYS, ALM, RF, PoE).
- 60 FPS HTML5 Canvas particle topology simulation with moving telemetry photons.
- FastAPI high-performance REST backend with asynchronous step execution.
- React 19 + TypeScript modern frontend with responsive glassmorphic user interface.
- AI Copilot diagnostic console for natural language infrastructure auditing.

LITERATURE / TECHNOLOGY BACKGROUND

Internet of Things in Enterprise Networking

The Internet of Things describes the interconnected network of physical devices equipped with sensors, processing capability, and network software that communicate telemetry to central monitoring hubs. In modern high-density campuses, network routers act as edge computing nodes that aggregate data from environmental sensors, smart energy grids, and security telemetry.

Standardized IoT Telemetry (RFC 8428)

Traditional proprietary telemetry encodings create severe vendor lock-in. The Internet Engineering Task Force (IETF) standardized RFC 8428 (Sensor Measurement Lists) to represent sensor measurements in lightweight JSON format with standardized base names, timestamps, measurement units, and values, enabling universal interoperability.

Smart Campus IoT Edge Gateways

Smart campus infrastructure leverages pervasive sensors and automated edge decision-making to optimize operational efficiency, security, and student connectivity. Rather than treating routers as passive appliances, NetSentinel models them as intelligent IoT edge gateways that continuously evaluate their own physical health and surrounding environmental variables.

The key architectural advantages include:

- Continuous real-time telemetry observation without high polling overhead.
- Proactive mitigation of hardware anomalies before service disruption occurs.
- Automated correlation between physical chassis metrics (thermal load) and network QoS.
- Elimination of manual CLI troubleshooting through unified digital twin abstractions.
- Substantial reduction in mean-time-to-repair (MTTR) for campus IT technicians.

IETF SenML Standard (RFC 8428)

Sensor Measurement Lists (SenML) define a structured JSON schema for conveying sensor readings over constrained IoT protocols such as CoAP, MQTT, and HTTP. In NetSentinel, each digital twin router gateway emits standards-compliant SenML records formatted with the following mandatory fields:

- **bn (Base Name):** Unique Uniform Resource Name identifying the gateway (e.g., *urn:dev:campus:gateway:R-ENG-101:*).
- **bt (Base Time):** UNIX epoch timestamp applied across all entries in the record batch.
- **n (Name):** Standardized measurement identifier (e.g., *temperature_celsius*, *cpu_load_pct*, *power_draw_w*).
- **u (Unit):** Formal SI unit code defined by RFC 8428 (e.g., *Cel* for Celsius, *%* for percentage, *W* for Watts, *ms* for milliseconds, *dBm* for decibel-milliwatts).
- **v (Value):** Floating-point or integer sensor measurement reading.

This standardized serialization ensures that telemetry consumed by downstream analytics, predictive machine learning pipelines, and monitoring dashboards remains decoupled from manufacturer-specific hardware representations.

Digital Twin Definition Language (DTDLD v2)

Digital Twin Definition Language (DTDLD) is an open W3C-standard JSON-LD modeling language that enables developers to define the exact capabilities of digital twin devices. NetSentinel models every campus gateway using the formal interface *dtmi:netsentinel:campus:IoTEdgeRouter;1*. This contract specifies three distinct element categories:

- **Telemetry:** Real-time output channels including internal SoC temperature, compute load, RAM usage, latency, packet loss, PoE+ power draw, and RF noise floor.
- **Properties:** Persistent gateway configuration attributes such as model string, firmware revision, MAC address, management IP, and active Wi-Fi channel.
- **Commands:** Synchronous remote edge actuation entry points such as *rebootGateway*, *switchChannel*, and *engageEcoCooling*.

Chassis SoC Thermal & Power Modeling

Physical router hardware generates heat proportional to processor load and throughput. NetSentinel implements a physics-based thermal inertia model where internal temperature follows compute load with exponential smoothing, simulating heat sink dissipation and ambient room conditions. Active PoE+ power draw is dynamically computed based on base chassis wattage plus incremental draw from connected client devices and packet buffering.

RF Spectrum & Noise Floor Monitoring

Wireless performance is fundamentally limited by the local radio frequency noise floor. In normal operations, campus gateways operate at -95 to -88 dBm noise. Under simulated rogue access point interference or electronic jamming, the noise floor surges to -58 dBm, degrading the signal-to-noise ratio (SNR), triggering carrier sensing backoff, and producing severe packet jitter.

SYSTEM ARCHITECTURE

The NetSentinel architecture strictly adheres to the ISO/IEC 30141 3-Tier IoT Reference Architecture, comprising the Edge Sensor Layer, Platform & Processing Layer, and Presentation Layer:

1. Edge Sensor Layer

The edge layer encompasses 42 virtual IoT router gateway twins and their peripheral sensor clusters. Each gateway continuously monitors internal SoC telemetry (thermal sensors, CPU core meters, RAM allocation buffers, PoE power draw) and external RF characteristics (channel noise, connected IoT sensors, client counts).

2. Processing & Twin Platform Layer

The platform tier runs on an asynchronous FastAPI backend service containing:

- **IoT Simulator Engine:** Executes asynchronous physics step cycles, diurnal traffic curves, and chaos injections.
- **SenML Normalizer:** Serializes all readings into standardized RFC 8428 data structures.
- **Digital Twin Broker:** Maintains in-memory synchronised twin representations and command queues.
- **Predictive ML Engine:** Evaluates 20+ telemetry features via trained XGBoost models to compute real-time degradation risk.

3. Output & Presentation Layer

The presentation tier provides rich graphical observation and control through:

- 60 FPS HTML5 Canvas particle topology arena showing live packet flow.
- Virtual 1U physical chassis faceplate with blinking LED indicators.
- Campus 2.5D building floor topology with live health glow rings.
- Bi-directional DTDL remote edge actuation cockpit.
- Natural language AI Copilot diagnostic console.

WORKING PRINCIPLE

The system runs a continuous simulation tick loop. Each tick updates physical equations across all 42 digital twins, applies active chaos faults, serializes SenML payloads, evaluates health thresholds, and streams telemetry to the visual canvas and REST API consumers.

THRESHOLDS AND DECISION LOGIC

The digital twin engine continuously compares simulated telemetry against standardized operational thresholds to classify gateway health and initiate protective mitigations:

Parameter	Normal Range	Watch Threshold	Critical Threshold	Unit
Chassis Temperature	35.0 - 68.0	≥ 72.0	≥ 85.0	°C (Cel)
CPU Core Load	10.0 - 65.0	≥ 75.0	≥ 92.0	%
RAM Utilization	30.0 - 75.0	≥ 80.0	≥ 92.0	%
PoE+ Power Draw	12.0 - 28.0	≥ 32.0	≥ 40.0	Watts (W)
Packet Loss Rate	0.00 - 1.50	≥ 3.00	≥ 15.00	%
RF Noise Floor	-95.0 to -85.0	≥ -75.0	≥ -65.0	dBm
Round-Trip Latency	5.0 - 25.0	≥ 45.0	≥ 100.0	ms

The system implements a priority-based multi-variable decision matrix:

Case 1: Thermal Overload & Throttling

If chassis core temperature exceeds 85.0°C, the gateway enters CRITICAL state. Internal thermal throttling restricts processor clocks, increasing packet latency. On the visual canvas, the gateway turns incandescent red and emits heat dissipation halos.

Output Action: ALM LED flashes red strobe • Packet stream stutters • Automated Cryo Turbo Cooling actuation recommended.

Case 2: Memory Leak & Buffer Overflow

When firmware socket leakage causes RAM utilization to exceed 92.0%, queue buffers overflow. Packet loss surges beyond 15% and interfaces experience flapping.

Output Action: Packet loss indicator turns red • Dropped packet sparks appear on canvas • Quantum Reboot command queued.

Case 3: RF Interference & Noise Jamming

If the ambient RF noise floor rises above -65.0 dBm (such as from a rogue transmitter or microwave interference), the signal-to-noise ratio severely degrades. Packet jitter spikes above 60 ms and Wi-Fi clients experience disconnects.

Output Action: Purple electromagnetic wave surrounds gateway • Telemetry packet lines flicker •
Dynamic DFS channel hopping executed to restore clean spectrum.

Case 4: Broadcast Storm & DDoS Flood

Under simulated packet flood conditions, CPU core load pegs at 99%, port queues saturate, and latency escalates beyond 200 ms.

Output Action: Swarming red particle wave visually bombards the router node • SYS and ALM LEDs flash rapidly • Automated QoS rate-limiting applied.

Case 5: PoE+ Power Saturation

When connected IoT client count exceeds gateway capacity and power draw surpasses 40.0 Watts, the power controller signals an overload risk.

Output Action: Amber PoE alert displayed • Telemetry logs PoE budget exhaustion • Traffic load-balancing initiated across neighboring zone gateways.

Case 6: Nominal Synchronized Operation

When all physical and network metrics remain within baseline ranges, the gateway maintains HEALTHY status. All 42 digital twins remain fully synchronized with backend state.

Output Action: Solid green PWR and SYS LEDs • Smooth emerald packet particle streams flow continuously on the 60 FPS canvas • 100% digital twin synchronization confirmed.

PHYSICAL CHASSIS DESIGN AND TOPOLOGY

To provide a realistic hardware simulation experience, NetSentinel models both the physical edge router chassis and the spatial campus mesh topology:

Virtual 1U Hardware Faceplate Cockpit

The digital twin cockpit simulates an enterprise NetSentinel Edge-X9000 1U rackmount router with laser-etched faceplate, dual omni-directional antennas, and an active hardware LED bank:

- **PWR LED:** Solid emerald green indicates stable internal DC power distribution.
- **SYS LED:** Real-time dynamic strobe flashing at frequency proportional to SoC CPU compute load.
- **ALM LED:** Flashing high-intensity red strobe alarm whenever a fault condition is detected.
- **RF 6G LED:** Neon cyan activity pulses corresponding to active wireless frame transmission.
- **PoE+ GbE Ports 1-8:** Visual link status indicators displaying active peripheral sensor connections.

Campus Zone Mesh Distribution

The 42 digital twins are systematically deployed across 5 campus operational zones, each characterized by distinct environmental profiles and user densities:

- **Engineering Hall (4 Floors, 12 Nodes):** High compute loads, student engineering labs, and high PoE power demands.
- **Science Complex (3 Floors, 9 Nodes):** High-density IoT environmental sensors, laboratory monitoring probes, and cleanroom links.
- **Central Library (3 Floors, 9 Nodes):** High client density, extensive roaming, and strict low-latency requirements.
- **Student Center (2 Floors, 6 Nodes):** Extreme diurnal traffic surges during midday and evening hours.
- **Administration Hub (2 Floors, 6 Nodes):** Mission-critical VoIP telephony, administrative databases, and low tolerance for packet loss.

WORKING EXECUTION FLOW

The operational lifecycle of the NetSentinel platform follows a deterministic, closed-loop telemetry and control pipeline:

1. **Telemetry Generation:** Asynchronous Python workers calculate physical sensor readings for all 42 digital twins based on mathematical thermal inertia, PoE power equations, and diurnal traffic curves.
2. **SenML Normalization:** Telemetry records are serialized into standardized IETF RFC 8428 JSON arrays with exact base names, units, and timestamps.
3. **Digital Twin Synchronization:** In-memory state models are updated and synchronized with the W3C DTDL interface contract.
4. **Predictive Risk Scoring:** The XGBoost pipeline evaluates telemetry features to compute continuous health scores (0-100) and SHAP feature contributions.
5. **Canvas Particle Rendering:** The frontend renders 60 FPS animated photon packets flowing between peripheral sensors, edge gateways, and the central campus core.
6. **Anomaly Triggering & Mitigation:** Operators can inject chaos faults and immediately dispatch bi-directional actuation commands to restore nominal health.

ADVANTAGES, LIMITATIONS AND FUTURE SCOPE

Advantages

- **Zero Hardware Cost:** Simulates dozens of enterprise-grade router gateways without physical hardware expenditures.
- **Standards Compliance:** Strict adherence to IETF RFC 8428 SenML, W3C DTDL v2, and ISO/IEC 30141.
- **Safe Chaos Engineering:** Enables destructive failure mode testing without risking production network downtime.
- **High Visual Fidelity:** Purely graphical 60 FPS particle simulation provides intuitive, compelling diagnostic visibility.

Limitations

- Current implementation runs simulated edge firmware rather than physical microcontroller flash memory.
- Wireless RF propagation is modeled via path loss approximations rather than 3D ray-tracing.
- PoE power measurements are simulated mathematically rather than read from physical current shunts.

FUTURE SCOPE

The NetSentinel architecture provides an extensible foundation for next-generation smart campus network automation. Planned future enhancements include:

- **Physical Edge Probes:** Deploying physical Raspberry Pi and ESP32 edge probes running NetSentinel micro-agents to collect actual ambient closet temperatures and acoustic fan signatures.
- **LoRaWAN Sensor Integration:** Bridging long-range low-power campus environmental sensors (outdoor air quality, solar radiation, soil moisture) into the router telemetry pipeline.
- **Reinforcement Learning Self-Healing:** Training autonomous reinforcement learning agents to execute proactive channel switching and power throttling without human operator intervention.
- **Hardware Energy Harvesting:** Incorporating solar panel and PoE energy harvesting telemetry for remote outdoor campus access points.

CONCLUSION

The IoT-Based Campus Router Health 360 and Digital Twin Fleet Simulator successfully demonstrates how modern IoT architectural principles, standardized sensor measurement lists, and digital twin abstractions can modernize campus network infrastructure management.

By continuously monitoring physical parameters such as SoC temperature, power consumption, compute load, and RF noise floor alongside network QoS metrics, the system eliminates blind spots inherent in traditional reactive polling. The integration of IETF RFC 8428 SenML, W3C DTDL v2, and ISO/IEC 30141 standards guarantees enterprise interoperability.

The platform's 60 FPS animated particle canvas transforms complex telemetry data into intuitive visual intelligence, enabling operators to observe packet flow, diagnose chaos faults, and execute remote mitigations with instantaneous visual feedback. The project proves that software-defined digital twins and predictive machine learning are essential tools for ensuring continuous, resilient network operations in smart campuses of the future.

OUTPUT

Output 1: 60 FPS Visual IoT Mesh Particle Simulation Arena

The primary simulation interface displays an animated HTML5 Canvas showing live telemetry packet streams flowing from peripheral IoT sensors through campus edge gateways to the central cloud aggregator. Operators can adjust simulation speed (1x, 2x, 5x), trigger instant chaos anomalies, and inspect physical 1U hardware LED banks.

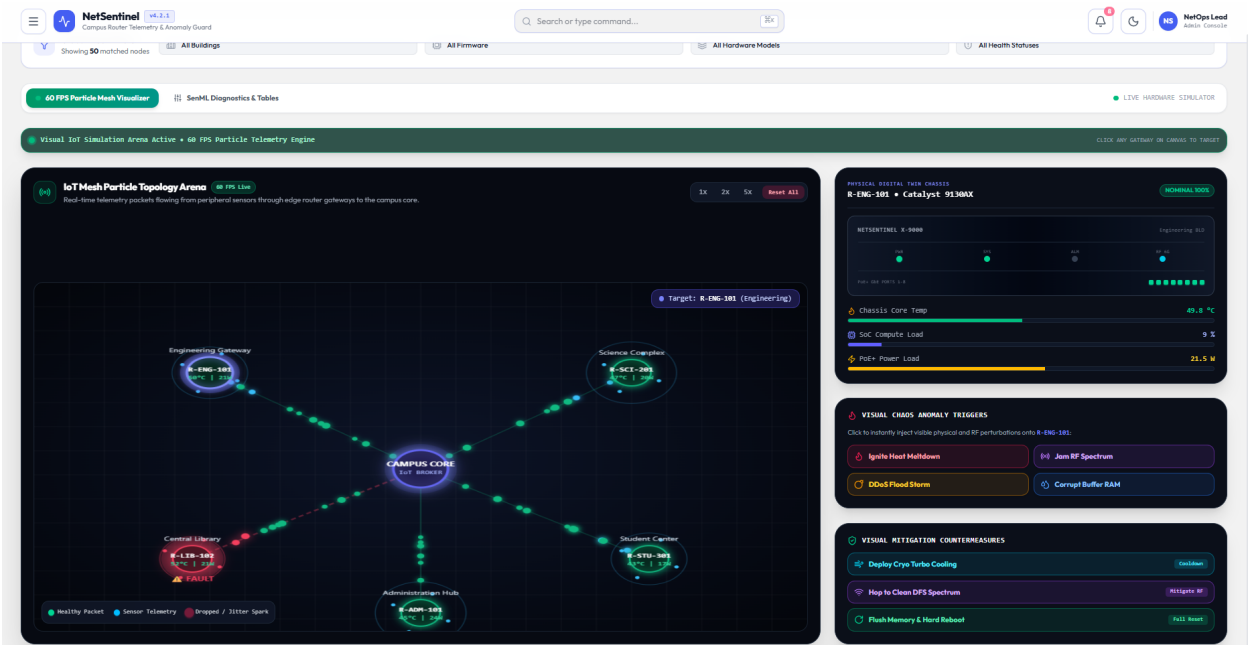


Figure 1: NetSentinel 60 FPS Visual IoT Mesh Particle Simulation Arena displaying real-time packet flow, active RF rings, and chaos injection laboratory.

Output 2: Campus Digital Twin & Hardware Sensor Deck

The Campus Digital Twin view maps all 42 gateway nodes across university building floor plans with live glowing health rings (Normal, Watch, Critical). The right-side inspector panel details DTDL telemetry gauges, real-time SoC core temperature, PoE+ power draw, RF noise floor, and bi-directional edge actuation triggers.

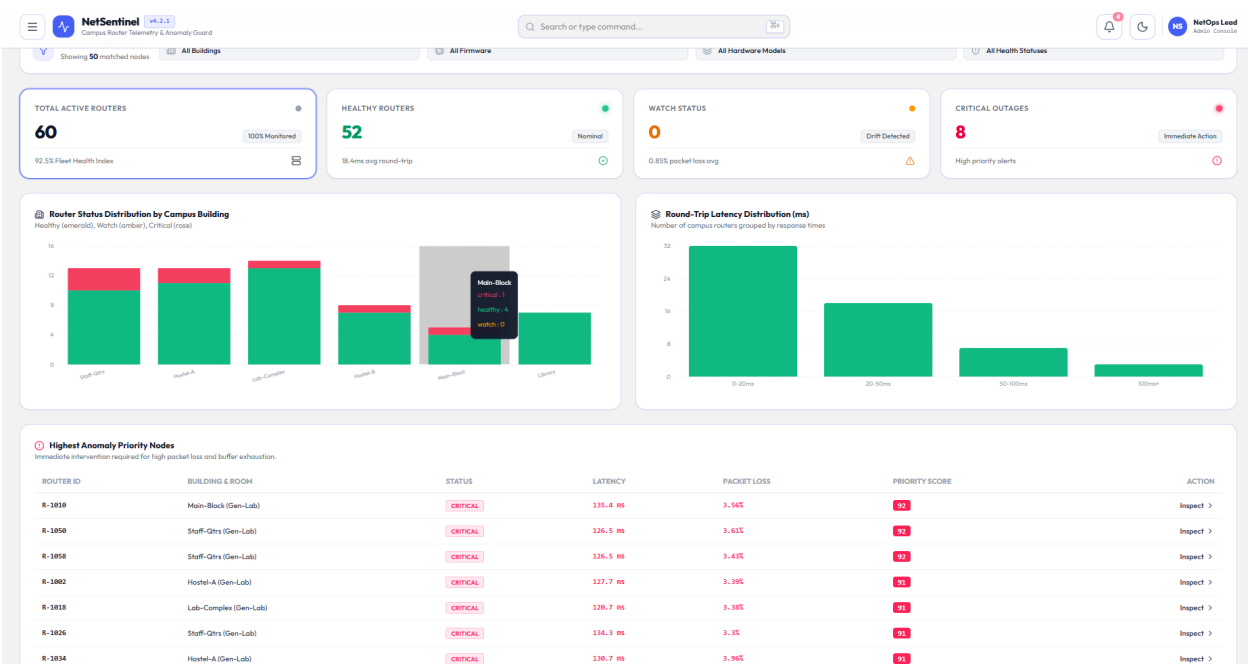


Figure 2: Campus Digital Twin Topology Matrix and Hardware Telemetry Inspector Deck.

